

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 1: Pitot-Static Systems & Temperature Measurement

Syllabus Source: CAE Oxford (pp. 1-38) | Pass Target: >= 90% AviationExam

1. PRESSURE SOURCING: PITOT PRESSURE & STATIC PRESSURE

Aeronautical pressure instruments (Altimeter, ASI, VSI, Machmeter) rely on samples of atmospheric ambient pressure and impact pressure captured through dedicated external fuselage sensors:

- Static Pressure (P_s): The ambient barometric pressure of undisturbed still air at the aircraft's flight level. Sampled via flush static vents mounted on opposite sides of the fuselage (plumbing joined in a balance pipe to cancel yaw/sideslip errors).
- Pitot Pressure (P_t / Total Pressure): The total stagnation pressure captured by open-ended, forward-facing pitot probes aligned parallel to the longitudinal axis: Total Pressure (P_t) = Static Pressure (P_s) + Dynamic Pressure (q).
- Dynamic Pressure (q): Pressure resulting from aircraft forward velocity: $q = \frac{1}{2} \rho V^2$. Pitot pressure alone cannot measure airspeed; the static pressure must be subtracted (P_t - P_s = q)!
- Alternate Static Source: Located inside unpressurized flight deck or avionics bay. Due to airflow venturi suction around the fuselage, pressure inside the cabin is LOWER than ambient outside static pressure! Selecting alternate static causes: Altimeter reads HIGH, ASI reads HIGH, VSI shows momentary climb then stabilizes.

2. PITOT-STATIC ERRORS & SENSOR HEATING

Measurement errors distort raw pressure readings before reaching instrument capsules:

Error Source	Physical Mechanism & Flight Effect	Correction & Operational Mitigation
Instrument Error	Mechanical imperfections, capsule hysteresis, and friction in gear linkages.	Calibrated during factory manufacturing; negligible in modern digital ADC air data sensors.
Position / Pressure Error	Airflow around fuselage disturbs ambient static pressure at the vent.	Varies with Angle of Attack and airspeed/flaps. Sensed by ADC and corrected electronically via lookup tables.
Maneuver / Lag Error	Time delay as pressure transmits through long pneumatic tubing.	Eliminated in digital aircraft by mounting ADMs (Air Data Modules) right beside the pitot/static probes.
Electrical Heating	MANDATORY electrical resistive heating elements inside pitot probes and static plates.	Prevents ice blockage. Toggled on automatically on engine start or flight crew checklist. Inoperative probe heater risks catastrophic instrument misreadings!

[!] EXAM TRAP (AVIATIONEXAM) - Alternate Static Source Selection Flight Deck Effects

When the alternate static valve is opened on unpressurized aircraft: - Cabin static pressure is LOWER than true ambient atmospheric pressure (fuselage Bernoulli suction). - Altimeter: Reads HIGHER than actual altitude! - Airspeed Indicator (ASI): Reads HIGHER than actual airspeed! - Vertical Speed Indicator (VSI): Shows a MOMENTARY FALSE CLIMB, then returns to indicate true rate of climb/descent!

3. AIR TEMPERATURE MEASUREMENT: SAT, TAT & RAM RISE

Accurate outside temperature is essential for True Airspeed (TAS) computation, Mach number calculation, and icing detection:

- Static Air Temperature (SAT / OAT): The true ambient temperature of undisturbed outside air through which the aircraft is flying.
- Total Air Temperature (TAT): The temperature measured by a stagnation probe (Rosemount probe) exposed to the airstream. Due to adiabatic kinetic compression and boundary layer friction, air is brought to rest at the probe sensor, generating kinetic heating ('Ram Rise'): $TAT = SAT + \text{Ram Rise}$.
- Ram Rise Formula: $\text{Ram Rise} = (V_{TAS} / 100)^2$, or thermodynamically: $TAT = SAT \times (1 + 0.2 \times K_r \times M^2)$, where temperatures are in Kelvin ($K = ^\circ C + 273$), M = Mach number, and K_r = probe recovery factor (~1.0 for flight test).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): SAT and Ram Rise Calculation**

Exam Problem: Cruising at Mach 0.80, indicated TAT is -10 C (263 K). Recovery factor $K_r = 1.0$. What is true static air temperature (SAT)? Step 1: Formula: $TAT = SAT \times (1 + 0.2 \times M^2)$. Step 2: Mach factor: $1 + 0.2 \times (0.80)^2 = 1 + 0.128 = 1.128$. Step 3: Solve for SAT in Kelvin: $SAT_K = 263 / 1.128 = 233.15 \text{ K}$. Step 4: Convert to Celsius: $SAT_C = 233.15 - 273.15 = -40 \text{ C}$. Rule: TAT is ALWAYS WARMER than SAT in flight due to ram compression.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In an unpressurized aircraft, what occurs to the primary flight instruments if the pilot selects the alternate

- [A]: Altimeter reads low, ASI reads low, VSI shows momentary descent
- [B]: Altimeter reads HIGH, ASI reads HIGH, and VSI shows a momentary false CLIMB
- [C]: All instruments read zero
- [D]: Only the Machmeter is affected

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Airflow over the fuselage creates suction inside the cockpit, making cockpit pressure lower than outside static. Lower static pressure causes the altimeter to indicate higher, the ASI differential to indicate faster, and the VSI to sense a momentary climb.

Q2: An aircraft is cruising at Mach 0.80 with a Static Air Temperature (SAT) of -50°C. What is the approximate T

- [A]: -70°C
- [B]: -50°C

- [C]: -21°C
- [D]: 0°C

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Formula: $TAT (K) = SAT (K) \times (1 + 0.2 \times M^2)$. Convert SAT: $-50^\circ C + 273 = 223 K$. Factor: $1 + 0.2 \times (0.80)^2 = 1 + 0.2 \times 0.64 = 1.128$. $TAT = 223 \times 1.128 = 251.5 K$. Convert back to Celsius: $251.5 - 273 = -21.5^\circ C (\sim -21^\circ C)$.

Q3: Which flight instruments require a pneumatic connection to the PITOT pressure line?

- [A]: Altimeter and VSI only
- [B]: Airspeed Indicator (ASI) and Machmeter only
- [C]: Altimeter, ASI and VSI
- [D]: Turn coordinator and artificial horizon

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Only the Airspeed Indicator and the Machmeter require pitot (total) pressure to measure dynamic pressure. The Altimeter and VSI connect exclusively to the static pressure line.

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Chapter 2: Pressure Altimeters & Vertical Speed Indicators

Syllabus Source: CAE Oxford (pp. 39-78) | Pass Target: >= 90% AviationExam

1. PRESSURE ALTIMETER MECHANICS & PRESSURE DATUMS

The pressure altimeter is an absolute pressure gauge calibrated to the ICAO Standard Atmosphere (ISA: MSL pressure 1013.25 hPa, temperature +15°C, lapse rate 1.98°C / 6.5°C per 1,000 m up to 36,090 ft):

- Internal Mechanism: Contains an EVACUATED aneroid capsule (sealed with internal near-perfect vacuum). Ambient static pressure enters the instrument case. As aircraft climbs, static pressure drops; the capsule expands under internal leaf-spring tension, rotating pointer needles via jewel-pivoted magnifying gears.
- QNH (Nautical Height): Altimeter subscale set to station barometric pressure adjusted to MSL using ISA temperature lapse. Reads aerodrome elevation when on the ground; indicates ALTITUDE above mean sea level in flight.
- QFE (Field Elevation): Subscale set to actual atmospheric pressure at aerodrome reference point. Reads ZERO on ground; indicates HEIGHT above airfield runway.
- Standard Setting (1013.25 hPa / 29.92 inHg): Set when climbing through the Transition Altitude (TA). Altimeter indicates FLIGHT LEVEL (FL). Standardizes vertical separation between all cruising traffic worldwide.

2. ALTIMETER TEMPERATURE & PRESSURE ERRORS

Error Type	Atmospheric Deviation Phenomenon	Mathematical Rule & Rule of Thumb
Error Type	Atmospheric Deviation Phenomenon	Mathematical Rule & Rule of Thumb
Pressure Error	Flying from high pressure into low pressure without updating QNH subscale.	True altitude is LOWER than indicated altitude! For every 1 hPa error: altimeter misreads by approximately 27 to 30 FEET (Rule: 1 hPa = 30 ft at sea level).
Temperature Error	Flying in air colder or warmer than standard ISA (+15°C at MSL, -2°C/1,000 ft).	Rule of Thumb: 4 ft per 1,000 ft of altitude for each 1°C of ISA deviation. Cold Air: True altitude is LOWER than indicated! Hot Air: True altitude is HIGHER than indicated!

[!] EXAM TRAP (AVIATIONEXAM) - Universal Altimeter Error Mnemonic

'HIGH TO LOW OR HOT TO COLD, LOOK OUT BELOW!' - When flying from high pressure to low pressure, or from warm air to cold air: - THE ALTIMETER READS ERRONEOUSLY HIGH! (You are dangerously closer to the ground/mountain peaks than the instrument indicates)! - In sub-zero winter temperatures, published IFR approach MDA/DA and minimum sector altitudes MUST be corrected using cold temperature correction tables!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Cold Temperature Altimeter Correction

Official EASA Rule: Correction = 4 ft x (Altitude / 1,000 ft) x (ISA deviation below standard). Problem: Flying at indicated altitude 8,000 ft over mountains. OAT at 8,000 ft is -25°C. What is true altitude? Step 1: Calculate ISA temp at 8,000 ft: 15°C - (2 x 8) = -1°C. Step 2: Deviation: -25°C - (-1°C) = 24°C colder than ISA. Step 3: Correction = 4 x 8 x 24 = 768 ft. Step 4: In cold air, altimeter overreads. True Altitude = 8,000 - 768 = 7,232 ft. Rule: 'High to Low, Hot to Cold, Look Out Below!'

3. STEP-BY-STEP WORKED EXAMPLE (FOR DUMMIES / PASO A PASO)**[*] OFFICIAL DEFINITION - PRACTICAL CALCULATION: Calculating Altimeter Temperature Error and True Altitude**

SCENARIO (AviationExam Classic Cold Weather Altimetry Drill): An aircraft is cruising at FL 100 with an indicated altitude of 10,000 ft on 1013.25 hPa: - The outside air temperature (OAT / SAT) is -25°C - The route passes over high mountain peaks with minimum clearance required QUESTION: Calculate the True Altitude of the aircraft, and assess the terrain clearance hazard. SOLUTION PASO A PASO (FOR DUMMIES): Step 1: Determine Standard ISA Temperature at 10,000 ft: - ISA Sea Level Temp = +15°C - ISA Lapse Rate = 2°C per 1,000 ft - ISA Temp at 10,000 ft = +15°C - (2°C x 10) = +15°C - 20°C = -5°C. Step 2: Calculate the ISA Temperature Deviation (Delta-ISA): - Actual Temp = -25°C - Delta-ISA = Actual - Standard = -25°C - (-5°C) = -20°C (Air is 20°C colder than standard ISA!). Step 3: Apply the 4 ft per 1,000 ft Rule of Thumb: - Formula: Temperature Correction = 4 ft x (Altitude / 1,000) x Delta-ISA - Correction = 4 x 10 x (-20) = -800 feet! Step 4: Calculate True Altitude: - True Altitude = Indicated Altitude + Correction - True Altitude = 10,000 ft - 800 ft = 9,200 ft! FINAL ANSWER: True Altitude is 9,200 ft. The aircraft is 800 ft lower than what the altimeter displays ('High to low, hot to cold, look out below!').

4. VERTICAL SPEED INDICATORS: VSI VS INSTANTANEOUS VSI (IVSI)

The Vertical Speed Indicator measures the rate of change of static pressure, indicating rate of climb or descent in feet per minute (fpm):

- Conventional VSI Mechanism: Contains a flexible diaphragm capsule fed directly with static pressure. The surrounding sealed case is fed through a calibrated capillary metering leak tube. In level flight, pressures equalize (0 fpm). In a climb, capsule pressure drops instantly while case pressure leaks slowly through the capillary, creating a differential pressure that deflects the needle.
- VSI Time Lag (6 to 9 Seconds): Because air takes several seconds to choke through the capillary leak, conventional VSIs suffer a 6 to 9 second lag before indicating a steady-state rate of climb/descent!
- Instantaneous VSI (IVSI): Eliminates time lag by incorporating inertia dashpot pistons (accelerometer pumps). When the aircraft enters a pitch change, inertial mass of the piston pumps instantaneous differential pressure into the capsule, giving ZERO-LAG vertical speed indication!

5. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: An aircraft is flying at FL 100 with an indicated altitude of 10,000 ft on 1013 hPa. Outside Air Temperature (OAT) is -25°C. What is the true altitude?

- [A]: 10,000 ft
- [B]: 10,800 ft
- [C]: 9,200 ft

- [D]: 8,500 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Standard ISA temp at 10,000 ft = $15 - (2 \times 10) = -5^{\circ}\text{C}$. Actual temp = -25°C -> ISA deviation = -20°C (COLD AIR). Correction formula: $4 \text{ ft} \times (\text{Alt} / 1,000) \times \text{Delta-ISA} = 4 \times 10 \times (-20) = -800 \text{ ft}$. True Altitude = $10,000 - 800 = 9,200 \text{ ft}$!

Q2: What is the primary operational advantage of an Instantaneous Vertical Speed Indicator (IVSI) over a conventional VSI?

- [A]: It requires no static pressure connection
- [B]: It eliminates the 6 to 9 second time lag by using inertia-activated dashpot accelerometer pistons
- [C]: It indicates airspeed during cruise
- [D]: It operates on 28V DC power

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Conventional VSIs suffer a 6-9 second pneumatic metering lag. An IVSI uses acceleration dashpot pistons that pump immediate differential pressure into the capsule during maneuvers, providing instant response.

Q3: What will the pressure altimeter indicate on the runway before take-off if the subscale is set to the aerodrome elevation above MSL?

- [A]: The aerodrome elevation above MSL
- [B]: Exactly ZERO feet
- [C]: Standard flight level zero
- [D]: Pressure altitude

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. QFE is station barometric pressure at the runway reference point. Setting QFE references the altimeter to field height, reading exactly 0 ft on the runway threshold.

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Chapter 3: Airspeed Indicators & The Machmeter

Syllabus Source: CAE Oxford (pp. 79-118) | Pass Target: >= 90% AviationExam

1. AIRSPEED INDICATOR (ASI) PRINCIPLES & AIRSPEED HIERARCHY

The Airspeed Indicator measures dynamic pressure ($q = \frac{1}{2} \rho V^2$). Pitot pressure feeds the inside of an expandable metal capsule, while static pressure feeds the sealed instrument case: Differential Pressure = $P_{\text{pitot}} - P_{\text{static}} = \text{Dynamic Pressure } (q)$:

- Indicated Airspeed (IAS): The raw airspeed reading displayed on the cockpit ASI dial or PFD speed tape, uncorrected for errors.
- Calibrated Airspeed (CAS / RAS): IAS corrected for instrument errors and position / pressure errors (static vent aerodynamic disturbance). CAS is equal to EAS at low sub-transonic airspeeds at sea level.
- Equivalent Airspeed (EAS): CAS corrected for atmospheric compressibility effects. Crucial at high speeds and altitudes (> 250 kt / > 10,000 ft) where compressed air molecules generate false excess dynamic pressure: $EAS = CAS - \text{Compressibility Correction}$.
- True Airspeed (TAS): The actual physical velocity of the aircraft relative to the surrounding air mass. Calculated by correcting EAS for actual ambient air density: $TAS = EAS \times \sqrt{\rho_0 / \rho}$. TAS increases with altitude for a constant IAS (~1.5% to 2% increase per 1,000 ft)!

2. ASI COLOR CODING ARCS (CS-23 / CS-25)

Color Arc / Line	Speed Range Limits	Aerodynamic Significance & Pilot Actions
Color Arc / Line	Speed Range Limits	Aerodynamic Significance & Pilot Actions
White Arc	V_{SO} to V_{FE}	Flap operating range. Lower limit is stall speed in landing configuration (V_{SO}); upper limit is maximum flap extended speed (V_{FE}).
Green Arc	V_{S1} to V_{NO}	Normal operating range. Lower limit is stall speed clean (V_{S1}); upper limit is Maximum Structural Cruising Speed (V_{NO}).
Yellow Arc	V_{NO} to V_{NE}	Caution range. Maneuvers allowed in smooth air only; do not make abrupt control movements.
Red Radial Line	V_{NE} (Never Exceed)	Never Exceed speed. Exceeding risks catastrophic structural flutter and aerodynamic breakup!

3. THE MACHMETER: MECHANISM & OPERATIONAL ERRORS

Mach number is the ratio of True Airspeed (TAS) to the local speed of sound (a): $M = TAS / a$. Local speed of sound depends EXCLUSIVELY on absolute temperature: $a = 38.94 \times \sqrt{T \text{ [Kelvin]}}$:

- Machmeter Architecture: Contains TWO separate capsules mechanically coupled via a floating sliding ratio divider arm: 1. Airspeed capsule (fed by pitot/static differential, sensing q); 2. Altimeter aneroid capsule (evacuated, sensing static ambient pressure P_s).
- Aeronautical Ratio: Because dynamic pressure q is proportional to $P_s \times M^2$, the ratio of $(P_t - P_s) / P_s$ depends ONLY ON MACH NUMBER! The altimeter capsule continuously modifies the pivot leverage of the airspeed capsule.

- Zero Temperature / Density Error: CRITICAL EXAM POINT: The Machmeter suffers NO DENSITY ERRORS and NO TEMPERATURE ERRORS! Temperature and density factors mathematically cancel out in the ratio of dynamic to static pressure. The Machmeter is subject ONLY to instrument error and position error!

[!] EXAM TRAP (AVIATIONEXAM) - Climbing / Descending at Constant Speed Scenarios

- CLIMB at Constant IAS: TAS INCREASES (density drops); Mach number INCREASES (temperature drops)! - CLIMB at Constant Mach: TAS DECREASES (until tropopause, then constant); IAS DECREASES rapidly! - CROSSOVER ALTITUDE: The altitude where the aircraft transitions from climbing at constant IAS to climbing at constant Mach number (typically FL 280 to FL 310).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Climb Rules for IAS vs Mach

Rule 1 (Climb at Constant IAS): Air density drops -> TAS increases to maintain dynamic pressure. Temperature drops -> speed of sound 'a' drops. Increasing TAS divided by decreasing 'a' means Mach number INCREASES. Rule 2 (Climb at Constant Mach): Above crossover altitude, climbing at constant Mach means dynamic pressure q must decrease as static pressure drops. Therefore IAS DECREASES. Summary: In climb: Constant IAS -> Mach climbs. Constant Mach -> IAS decays.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: What errors affect a mechanical aircraft Machmeter?

- [A]: Density error and temperature error
- [B]: Instrument error and position (pressure) error ONLY
- [C]: Compressibility error only
- [D]: Lag error only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Machmeter measures the ratio of dynamic pressure to static pressure. Air density and temperature variables mathematically cancel out. Therefore, a Machmeter has NO density or temperature errors!

Q2: As an aircraft climbs at a constant Indicated Airspeed (IAS) of 280 knots from sea level up to FL 280, what

- [A]: Both TAS and Mach number decrease
- [B]: Both TAS and Mach number INCREASE
- [C]: TAS increases while Mach number remains constant
- [D]: TAS remains constant while Mach number increases

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. As altitude increases, air density drops, so TAS must increase for the same dynamic pressure (IAS). Simultaneously, outside air temperature drops, lowering the local speed of sound; therefore, Mach number increases rapidly!

Q3: What is the aerodynamic definition of Equivalent Airspeed (EAS)?

- [A]: IAS corrected for density altitude

- [B]: Calibrated Airspeed (CAS) corrected for compressibility effects
- [C]: Groundspeed corrected for wind
- [D]: True airspeed at the tropopause

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Equivalent Airspeed represents Calibrated Airspeed corrected for the adiabatic compressibility of air molecules at high forward speeds and high altitudes.

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Chapter 4: Air Data Computers (ADC) & ADIRU Architecture

Syllabus Source: CAE Oxford (pp. 119-148) | Pass Target: >= 90% AviationExam

1. AIR DATA COMPUTER (ADC) INPUTS & COMPUTED OUTPUTS

Modern glass-cockpit airliners replace individual mechanical instrument capsules with centralized digital Air Data Computers (ADC):

ADC Functional Category	Aeronautical Parameters & Connected Systems
Raw Sensor Inputs to ADC	Pitot Pressure (P _t), Static Pressure (P _s), Total Air Temperature (TAT), Angle of Attack (Alpha).
Primary Computed Outputs	Pressure Altitude (1013.25 hPa), Baro-corrected Altitude (QNH/QFE), Calibrated Airspeed (CAS), True Airspeed (TAS), Mach Number, Static Air Temperature (SAT), Vertical Speed (VSI).
Downstream Consumer Systems	EFIS Primary Flight Displays, Autopilot (AFCS), Flight Management System (FMS), TCAS II, GPWS/EGPWS, FADEC engine controllers, and Transponder Mode S Altitude Reporting.

- Electronic Position Error Correction (SSEC): The ADC automatically applies Static Source Error Corrections (SSEC) stored in digital calibration tables based on current Mach number, AOA, and flap position, achieving flawless altitude and speed precision.

2. AIR DATA INERTIAL REFERENCE UNIT (ADIRU) ARCHITECTURE

Modern commercial airliners (e.g. Airbus A320/A350/A380, Boeing 777/787) integrate the ADC and the Inertial Reference System (IRS) into a single unified line replaceable unit called an ADIRU:

- ADIRU Segregation: Each ADIRU contains two distinct, fault-isolated parts: 1. The ADR (Air Data Reference) part, processing pitot/static/temp data; 2. The IR (Inertial Reference) part, processing laser gyro and accelerometer attitude/position data.
- Triple Redundancy: Transport category aircraft install three independent ADIRUs (ADIRU 1, ADIRU 2, ADIRU 3). ADIRU 1 supplies the Captain's PFD/ND; ADIRU 2 supplies the First Officer's PFD/ND; ADIRU 3 acts as a standby hot backup selectable via cockpit switching.
- ADIRU Rotary Mode Selector: OFF: Unit depowered; NAV: Normal operating mode (inertial navigation active); ATT: Attitude-only reversion mode (used if navigation position computation fails in flight; provides basic pitch, roll, and heading, but velocity/position navigation is lost!).

[!] EXAM TRAP (AVIATIONEXAM) - ADIRU Switching & In-Flight Reversion
- If an ADR fault occurs: The pilot switches the Air Data selector to CAPT ON 3 or F/O ON 3. - If an IR fault occurs in flight: The pilot switches the mode selector to ATT (Attitude mode). The aircraft MUST be flown straight and level for 30 seconds to level the attitude platform. Entering magnetic heading manually via MCDU is mandatory in ATT mode!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Air Data Switching Protocol

Scenario: Unreliable airspeed alert on Captain PFD due to pitot icing on Probe

1. Step 1: Cross-check Captain PFD, First Officer PFD, and Standby ISFD. Step 2: On overhead AIR DATA switching panel, turn Captain selector from 'NORM' to 'CAPT ON 3'. Step 3: Captain PFD now receives valid air data from standby ADIRU 3, clearing the flag.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Which of the following parameters is directly COMPUTED by an Air Data Computer (ADC) rather than sensed?

- [A]: Pitot pressure
- [B]: Total Air Temperature
- [C]: True Airspeed (TAS) and Mach Number
- [D]: Angle of attack

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Pitot pressure, static pressure, TAT, and AOA are raw physical inputs. True Airspeed, Mach number, and calibrated airspeed are mathematically computed by the ADC processors.

Q2: In an aircraft equipped with three ADIRUs, what capability remains if an Inertial Reference (IR) unit is switched off?

- [A]: Full LNAV and RNAV navigation
- [B]: Pitch, roll attitude, and heading information ONLY (navigation capability is lost)
- [C]: Autoland CAT III capability
- [D]: TCAS resolution advisories

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The ATT (Attitude) mode is an emergency reversion. It retains basic gyro attitude (pitch/roll) and magnetic heading, but completely disables inertial position, groundspeed, and navigation tracking.

Q3: What is the operational purpose of Static Source Error Correction (SSEC) within an Air Data Computer?

- [A]: To de-ice the pitot probe
- [B]: To automatically compensate for aerodynamic position errors around the static vents across various Mach numbers and angles of attack
- [C]: To calibrate cockpit compasses
- [D]: To measure fuel density

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Static vents suffer position error caused by airflow pressure disturbances. The ADC applies SSEC algorithms to eliminate position error, ensuring highly accurate altitude and airspeed readouts.

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Chapter 5: Terrestrial Magnetism & Direct-Reading Compass

Syllabus Source: CAE Oxford (pp. 149-188) | Pass Target: $\geq 90\%$ AviationExam

1. TERRESTRIAL MAGNETISM, VARIATION & DIP ANGLE

The Earth behaves like a giant bar magnet with its magnetic South pole located near geographic North. Aircraft magnetic navigation relies on three fundamental terrestrial parameters:

- **True North vs Magnetic North:** True North is the geographic rotational axis of the Earth. Magnetic North is the direction indicated by a freely suspended magnetic needle aligned with terrestrial flux lines.
- **Magnetic Variation (Declination):** The angular difference between True North and Magnetic North. Varies geographically: Variation West (Magnetic is West of True: $\text{True} = \text{Mag} - \text{Var}$); Variation East (Magnetic is East of True: $\text{True} = \text{Mag} + \text{Var}$). Mnemonic: 'East is Least (-), West is Best (+)'.
- **Isogonals & Agonic Line:** Isogonals are lines joining points of equal magnetic variation. The Agonic line is the line of ZERO variation.
- **Magnetic Dip (Inclination):** The angle between the Earth's magnetic flux lines and the horizontal local plane. At the Magnetic Equator (Aclinic line), $\text{Dip} = 0^\circ$ (purely horizontal). At the Magnetic Poles, $\text{Dip} = 90^\circ$ (purely vertical). Magnetic compasses become completely useless near the magnetic poles!

2. DIRECT-READING COMPASS TURNING & ACCELERATION ERRORS

Direct-reading magnetic compasses use a pendulously suspended magnet system whose center of gravity lies below the pivot point. Due to magnetic dip, the compass suffers severe turning and acceleration errors in the Northern Hemisphere:

Compass Error Scenario	Physical Mechanism (Inertia + Magnetic Dip)	Pilot Operational Correction Rule
Acceleration Errors (East/West Headings)	Accelerate -> Inertia holds weight back -> Card tilts -> Dip pulls needle NORTH! Decelerate -> Card tilts opposite -> Needle pulls SOUTH.	Mnemonic: ANDS (Accelerate North, Decelerate South). On East or West headings in Northern Hemisphere!
Turning Errors (North/South Headings)	Turning THROUGH North: Centrifugal force tilts card outwards; dip pulls needle in turn direction -> Compass LAGS or turns wrong way! Turning THROUGH South: Dip pulls needle ahead of turn -> Compass LEADS (spins faster than aircraft).	Mnemonic: UNOS (Undershoot North, Overshoot South). Roll out BEFORE reaching North; roll out PAST South!

[!] EXAM TRAP (AVIATIONEXAM) - Northern Hemisphere Compass Errors Rules

- ANDS (Acceleration): When accelerating on East or West heading -> Compass indicates a turn towards NORTH! When decelerating -> Compass indicates a turn towards SOUTH! - UNOS (Turning): When turning to a Northerly heading -> Undershoot roll-out (roll out $\sim 20\text{-}30^\circ$ before heading). When turning to a Southerly heading -> Overshoot roll-out (roll out $\sim 20\text{-}30^\circ$ past heading)! - In the Southern Hemisphere: Acceleration and turning errors are EXACTLY REVERSED (SAND / ONUS)!

3. AIRCRAFT DEVIATION & COMPASS SWING

Aircraft metals and electrical wiring generate local magnetic fields that deflect the compass needle away from Magnetic North to Compass North:

- Deviation: The angular difference between Compass Heading and Magnetic Heading: Magnetic = Compass +/- Deviation.
- Compass Swing: A calibrated ground maintenance procedure conducted on a certified compass rose to align permanent corrector magnets, minimizing residual deviation. Residual errors are recorded on the flight deck Compass Deviation Card (maximum residual deviation $\leq 5^\circ$).

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): ANDS and UNOS Compass Error

Rule ANDS (East/West Headings): - Accelerate -> Compass turns NORTH. - Decelerate -> Compass turns SOUTH. Rule UNOS (Turning Headings in Northern Hemisphere): - Turning to NORTH: Compass leads/skips ahead -> Roll out EARLY (Undershoot North). - Turning to SOUTH: Compass lags behind -> Roll out LATE (Overshoot South).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In the Northern Hemisphere, what does a direct-reading magnetic compass indicate when the aircraft ACC

- [A]: A turn towards South
- [B]: A turn towards NORTH
- [C]: No change in heading
- [D]: Rapid 360-degree rotation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Mnemonic ANDS: Accelerate North, Decelerate South. Accelerating on an East or West heading tilts the pendulous card, allowing the vertical component of Earth's magnetic field (dip) to deflect the card towards North.

Q2: When making a turn onto a heading of NORTH (360°) in the Northern Hemisphere, how should the pilot roll

- [A]: Exactly when the compass passes 360°
- [B]: Roll out BEFORE reaching 360° (Undershoot North), because the compass lags behind the turn
- [C]: Roll out past 360° (Overshoot North)
- [D]: Increase turn rate to Rate 2

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Mnemonic UNOS: Undershoot North, Overshoot South. When turning through North in the Northern Hemisphere, compass liquid drag and dip cause the card to lag behind the turn, requiring an early rollout.

Q3: What is the name of the line on an aeronautical chart connecting points of ZERO magnetic variation?

- [A]: Isogonic line
- [B]: Agonic line
- [C]: Aclinic line
- [D]: Rhumb line

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An isogonal connects points of equal variation; the Agonic line is the specific isogonal of zero magnetic variation (where True North and Magnetic North coincide).

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 6: Gyroscopic Principles & Basic Gyro Instruments

Syllabus Source: CAE Oxford (pp. 189-228) | Pass Target: $\geq 90\%$ AviationExam

1. FUNDAMENTAL PROPERTIES OF GYROSCOPES

A mechanical gyroscope consists of a heavy symmetrical rotor spinning at high RPM (typically 12,000 to 24,000 RPM) mounted in gimbal rings providing degrees of freedom:

- **Rigidity in Space (Gyroscopic Inertia):** The property of a spinning rotor to resist any force attempting to alter its plane of rotation in space (Newton's 1st Law). Directly proportional to: 1. Rotor Mass (m); 2. Radius of gyration (r); 3. Rotational Speed (omega): Angular Momentum $L = I \times \omega$.
- **Gyroscopic Precession:** When an external torque (force) is applied to deflect the spin axis, the rotor resists in the direction of force and instead tilts at a point 90° LATER in the direction of rotation! Precession rate is inversely proportional to rotor speed.

2. CLASSIC GYROSCOPIC INSTRUMENTS: DI, AI & TURN COORDINATOR

Instrument	Rotor Spin Axis & Degrees of Freedom	Operating Function & Real/Apparent Errors
Instrument	Rotor Spin Axis & Degrees of Freedom	Operating Function & Real/Apparent Errors
Directional Gyro (Heading Indicator)	HORIZONTAL spin axis; 2 degrees of freedom (tied to azimuth).	Indicates aircraft heading. Free gyro: suffers Apparent Drift due to Earth's rotation: $\text{Drift} = 15^\circ/\text{hour} \times \sin(\text{Latitude})$. At 60°N: $15 \times \sin(60^\circ) = 13^\circ/\text{hour}$ apparent drift. Must be reset to magnetic compass every 10-15 min!
Artificial Horizon (Attitude Indicator)	VERTICAL spin axis; 2 degrees of freedom (tied to vertical).	Indicates Pitch and Roll attitude. Erected by pendulous vane air jets or electrical torque motors. Acceleration errors (indicates false climb and right roll on take-off acceleration); Turning errors (indicates false climb and opposite bank on rolling out of 180° turn).
Turn & Slip Indicator (Turn Coordinator)	HORIZONTAL spin axis; ONE degree of freedom with calibrated restraining springs.	Rate gyro: rate of yaw precesses gimbal against spring, deflecting needle. Rate 1 turn = 3°/sec (360° turn in 2 minutes). Slip ball (inclinometer) measures apparent gravity/centrifugal balance (shows slip or skid).

[!] EXAM TRAP (AVIATIONEXAM) - Apparent Drift (Earth Rate Wander) Calculation Formula

- Formula: $\text{Apparent Wander} = 15^\circ \times \sin(\text{Latitude})$ per hour! - At the Equator (Lat 0°):
 $\text{Drift} = 15 \times \sin(0^\circ) = \text{ZERO drift!}$ - At the North Pole (Lat 90°): $\text{Drift} = 15 \times \sin(90^\circ) = 15^\circ$ PER HOUR clockwise! - Direction: In the Northern Hemisphere, a free directional gyro drifts to the RIGHT (clockwise); in the Southern Hemisphere, it drifts to the LEFT!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Gyro Earth Rate Apparent Drift

Formula: Drift = $15 \text{ deg} \times \sin(\text{Latitude})$ per hour. Example at 60 N: $\sin(60) = 0.866 \rightarrow$

Drift = $15 \times 0.866 = 13 \text{ deg/h}$ to the right. At Equator (0 deg): $\sin(0) = 0 \rightarrow$ Drift is

ZERO. At Poles (90 deg): $\sin(90) = 1.0 \rightarrow$ Drift is MAXIMUM (15 deg/h).

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: A free directional gyroscope is operated at Latitude 60° North. What is the calculated apparent drift rate due to Earth's rotation?

- [A]: 0° per hour
- [B]: 7.5° per hour
- [C]: 13.0° per hour
- [D]: 15.0° per hour

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Formula: Earth Rate Drift = $15^\circ \times \sin(\text{Lat})$ per hour = $15^\circ \times \sin(60^\circ) = 15 \times 0.866 = 12.99^\circ$ per hour ($\sim 13.0^\circ/\text{hr}$).

Q2: In a basic air-driven artificial horizon (attitude indicator), what false indication occurs when the aircraft accelerates forward?

- [A]: A false nose-down pitch
- [B]: A false pitch-UP and a slight roll to the RIGHT
- [C]: A false rapid turn to the left
- [D]: No error occurs

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Take-off forward acceleration closes the rear pendulous erection vanes under inertia, causing the gyro to precess, generating a false indication of pitch-up and a slight false bank to the right.

Q3: How many degrees of freedom does the rate gyro inside a Turn and Slip indicator possess?

- [A]: One degree of freedom (gimbal constrained by springs)
- [B]: Two degrees of freedom
- [C]: Three degrees of freedom
- [D]: Zero degrees of freedom

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [A]. A rate gyro has only ONE gimbal ring (one degree of freedom), restrained by calibrated mechanical centering springs. Gyro precession directly measures the rate of angular yaw.

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Chapter 7: Inertial Reference Systems & Ring Laser Gyros

Syllabus Source: CAE Oxford (pp. 229-268) | Pass Target: >= 90% AviationExam

1. STRAPDOWN IRS ARCHITECTURE VS GIMBALLED INS

Inertial systems calculate present position, velocity, and attitude completely autonomously without receiving external radio signals or satellites:

- Gimballed INS (Older Technology): Mounted on a stabilized physical gimbal platform kept level with local Earth horizontal by torque motors. Heavy, mechanically complex, prone to gimbal lock during aerobatic maneuvers.
- Strapdown IRS (Modern Commercial Standard): Three Ring Laser Gyros (RLGs) and three solid-state accelerometers are rigidly 'strapped down' directly to the aircraft fuselage frame. A high-speed digital navigation computer mathematically converts body-axis accelerations into Earth-referenced horizontal navigation coordinates.

2. RING LASER GYRO (RLG) & THE SAGNAC EFFECT

Ring Laser Gyros have NO MOVING MECHANICAL ROTORS. Rotation is sensed using laser light optics via the Sagnac Effect:

- Operating Principle: Two monochromatic helium-neon laser beams travel in opposite directions (clockwise and counter-clockwise) around a sealed triangular cavity of Zerodur glass-ceramic via dielectric mirrors.
- The Sagnac Effect: When the aircraft rotates about the gyro axis, the path length for the beam traveling in the direction of rotation INCREASES, while the path length for the counter-rotating beam DECREASES. This path difference creates an optical frequency shift: $\Delta f = (4 \times \text{Area} \times \Omega) / (\text{Wavelength} \times \text{Optical Perimeter})$.
- Interference Fringe Detection: The two beams combine at a photodetector, creating optical interference fringes. The fringe transit rate is directly proportional to angular rotation rate.
- Laser Lock-In & Piezo-Dither Motor: At very low rotation rates ($< 0.1^\circ/\text{sec}$), backscattering causes the two laser frequencies to lock together, producing ZERO output (dead band). To prevent lock-in, a piezo-electric dither motor vibrates the RLG back and forth mechanically at ~400 Hz through the lock-in zone!

3. IRS ALIGNMENT PHASE & SCHULER TUNING

Phase / Principle	Operational Process & Duration	Flight Crew Action & Mathematical Feature
Phase / Principle	Operational Process & Duration	Flight Crew Action & Mathematical Feature
Initial Alignment (Stationary)	Takes 5 to 10 minutes (longer at high latitudes, e.g. 15 min at 70°N). AIRCRAFT MUST REMAIN COMPLETELY STATIONARY! (Cargo loading or wind buffeting will abort alignment).	Crew inputs present gate coordinates into MCDU. IRS measures Earth's rotation (15°/hr) to find TRUE NORTH (gyrocompassing) and senses gravity to level accelerometers.
Latitude Limitations	IRS cannot align above Latitude 78° or 82° North/South.	At extreme polar latitudes, the horizontal component of Earth rotation is too weak for gyrocompassing.

Schuler Tuning (84.4 Minutes)

Accelerometers are mathematically tuned to oscillate like an 84.4-minute pendulum whose length equals Earth radius ($R = 6,371 \text{ km}$).

Prevents aircraft acceleration from being mistaken for a tilt of the vertical. Schuler tuning contains navigation position errors within a periodic 84.4-minute bounded oscillation cycle!

[!] EXAM TRAP (AVIATIONEXAM) - IRS Alignment Rules & Lat/Long Verification

- The aircraft MUST NOT BE MOVED during the alignment cycle! - The flight crew must enter present position coordinates into the MCDU. The IRS compares entered latitude with its own internally sensed gyrocompass latitude: if discrepancy $> 1^\circ$, alignment FAILS! - Note: IRS navigates referenced to TRUE NORTH, not Magnetic North!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): IRS Alignment & Sagnac Effect**

Step 1: Aircraft must be 100% stationary on ramp. Step 2: RLG sensors detect Earth rotation ($15 \text{ deg/h} \times \cos(\text{latitude})$) to find True North automatically. Step 3: Accelerometers sense gravity vector to level the platform. Step 4: If entered gate position differs by >1 degree latitude, ALIGN annunciator flashes.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the physical optical phenomenon utilized by a Ring Laser Gyro (RLG) to measure angular rotation**

- [A]: Doppler effect
- [B]: The Sagnac effect, where optical path lengths differ between counter-propagating laser beams
- [C]: Photo-electric emission
- [D]: Faraday induction

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Sagnac effect dictates that rotating the laser cavity creates a path length difference between the clockwise and counter-clockwise beams, producing a frequency difference proportional to rotation rate.

Q2: What is the purpose of the piezo-electric 'dither motor' installed on a Ring Laser Gyro?

- [A]: To cool the laser mirrors
- [B]: To vibrate the gyro back and forth through the low-speed 'lock-in' threshold, preventing zero-output error
- [C]: To level the platform
- [D]: To align with magnetic north

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. At low rotation rates, internal mirror backscattering locks the two laser frequencies together (laser lock-in). A dither motor mechanically oscillates the gyro at high frequency to keep it out of the dead band.

Q3: What is the period of the Schuler oscillation cycle in an inertial reference navigation system?

- [A]: 24.0 hours
- [B]: 84.4 minutes
- [C]: 12.5 minutes
- [D]: 360 seconds

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The Schuler pendulum period is $T = 2 \times \pi \times \sqrt{R / g} = 84.4$ minutes.

Tuning the IRS to 84.4 minutes isolates linear accelerations from gravity vectors, preventing unbounded position drift.

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Chapter 8: Low-Altitude Radio Altimeter (FMCW)

Syllabus Source: CAE Oxford (pp. 269-298) | Pass Target: >= 90% AviationExam

1. FREQUENCY MODULATED CONTINUOUS WAVE (FMCW) RADAR

The Radio Altimeter (Rad Alt) measures the precise absolute geometric clearance of aircraft main landing gear wheels above underlying terrain from 0 to 2,500 ft AGL:

- Frequency Band: Operates in the SHF aeronautical radionavigation band between 4.2 GHz and 4.4 GHz (typically center frequency 4.3 GHz).
- FMCW Modulation Technique: Instead of high-power pulses, the transmitter outputs a CONTINUOUS signal whose carrier frequency is swept linearly up and down at a constant modulation rate (triangular or sinusoidal frequency sweep).
- Beat Frequency Principle: Signal transmitted downward reflects off the terrain and returns to the receiver antenna after a time delay: $\Delta t = (2 \times \text{Height}) / c$. During this transit time, the transmitter has swept to a new frequency! Mixing the received signal with the current transmitted signal produces a BEAT FREQUENCY (Δf) that is DIRECTLY PROPORTIONAL to terrain wheel height!

2. INTEGRATION, DECISION HEIGHT & AUTOLAND GUIDANCE

Height Regime	Operational Function & Cockpit Indications	Interfaced Aircraft Systems
Height Regime	Operational Function & Cockpit Indications	Interfaced Aircraft Systems
Above 2,500 ft AGL	Radio altimeter indication is INHIBITED or flagged OFF. Pointer parked behind mask; digital readout blanked.	No ground interaction; GPWS radar terrain clearance modes inactive.
2,500 ft down to 50 ft	Readout appears on PFD (round dial or digital tape). Dial expansion: 0-500 ft scale is expanded for high visual precision.	Feeds GPWS Modes 1-4; arms Autoland flare and rollout modes.
Decision Height (DH)	Pilot dials DH bug on EFIS control panel (e.g. 200 ft for CAT I, 100 ft or 50 ft for CAT II).	At DH: Yellow 'DH' annunciates on PFD and synthetic voice announces 'DECISION' or 'MINIMUMS'!
Below 50 ft (Autoland)	Crucial input to autopilot autoland computers.	At 50-30 ft: Triggers autopilot FLARE mode (throttles retard to idle, pitch increases smoothly for touchdown). Synthetic voice callouts: '50, 40, 30, 20, 10, RETARD'.

[!] EXAM TRAP (AVIATIONEXAM) - Radio Altimeter Zero Height Calibration

- The Radio Altimeter antennas are mounted on the fuselage underside. - Calibration Datum: In a level flight landing attitude, when the main landing gear tyres touch the runway concrete, the radio altimeter MUST INDICATE EXACTLY ZERO FEET! - Antenna cable delay lines are built into the receiver harness to compensate for the physical distance between antennas and main wheels.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Radio Altimeter vs Pressure Altimeter**

Rule 1: Pressure altimeter measures static pressure -> Altitudes (MSL) and Flight Levels. Rule 2: Radio altimeter transmits FMCW at 4.2-4.4 GHz -> Absolute height above ground (AGL). Rule 3: Radio altimeter operates ONLY between 0 and 2,500 ft AGL. Above 2,500 ft, pointer is parked or flag displays.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What modulation principle is utilized by a modern low-altitude radio altimeter?**

- [A]: Amplitude Modulation (AM)
- [B]: Frequency Modulated Continuous Wave (FMCW) where the beat frequency between transmitted and reflected waves is proportional to wheel height
- [C]: Pulsed primary radar with 10 microsecond pulses
- [D]: Phase shift keying

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Low-altitude radio altimeters use FMCW. Comparing the transmitted swept frequency against the delayed reflected return frequency produces a beat frequency directly proportional to terrain height.

Q2: At what radio altitude does the radio altimeter indication normally become active on a transport aircraft Prior to landing?

- [A]: 10,000 ft AGL
- [B]: 5,000 ft AGL
- [C]: 2,500 ft AGL
- [D]: 500 ft AGL

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Commercial radio altimeters operate from 0 to 2,500 ft AGL. Above 2,500 ft, the indication is blanked or masked on modern electronic displays.

Q3: When an aircraft is in the landing flare with its main gear wheels touching the runway surface, what should the radio altimeter display?

- [A]: Negative 5 ft
- [B]: Exactly ZERO feet
- [C]: Antenna height (approx 10 ft)
- [D]: Aerodrome elevation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Radio altimeters are calibrated with an internal electronic delay line so that with landing gear extended in touchdown attitude, the display reads exactly 0 ft when the tyres touch down.

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Chapter 9: Electronic Flight Instrument Systems (PFD & ND)

Syllabus Source: CAE Oxford (pp. 299-348) | Pass Target: >= 90% AviationExam

1. EFIS ARCHITECTURE: DISPLAY UNITS & SYMBOL GENERATORS

Electronic Flight Instrument Systems (EFIS) replace electromechanical gauges with high-resolution digital screens driven by central graphic processing computers (Display Computers / Symbol Generators - SG):

- Screen Types: Older airliners use Cathode Ray Tubes (CRT). Modern aircraft use Active Matrix Liquid Crystal Displays (AMLCD) providing wider viewing angles, lower power consumption, and daylight readability.
- Display Redundancy & Reversion: Each pilot seat has a Primary Flight Display (PFD) and a Navigation Display (ND). If a PFD screen burns out or fails, an EFIS display transfer switch moves the entire PFD image automatically or manually onto the adjacent ND screen!

2. STANDARD EFIS COLOR CODE PHILOSOPHY

EASA CS-25 certification standardizes EFIS color usage across all manufacturers to prevent pilot misinterpretation:

EFIS Color Standard	Aeronautical Meaning & Cockpit Applications
RED	Warnings, immediate flight hazards, flight envelope red line limits (e.g. V_NE, V_MO / M_MO barber pole, stick shaker speed band, TCAS RA red blocks).
AMBER / YELLOW	Cautions, abnormal conditions, armed limit ranges, airspeed caution range (yellow arc), TCAS TA advisory circles.
GREEN	Engaged active autopilot/flight director modes on the FMA, active navigation tracking (VOR/ILS course line), normal system status.
MAGENTA	Active flight guidance targets! Flight Director steering command bars, active FMS route flight plan, active target altitude, active waypoint name, VNAV descent path.
CYAN (LIGHT BLUE)	Armed autopilot modes on the FMA (waiting for capture), pre-selected bug values, secondary flight routes, tuned navigation aids.
WHITE	Current active numerical flight data, instrument dials and scales, non-active flight plan waypoints, system baseline borders.

3. PRIMARY FLIGHT DISPLAY (PFD) & NAVIGATION DISPLAY (ND) MODES

The PFD centralizes the classic 'Basic T' instruments on a single screen, topped by the Flight Mode Annunciator (FMA):

- Flight Mode Annunciator (FMA): Divided into 5 columns across the top of the PFD: 1. Autothrottle status; 2. Vertical mode (engaged green / armed cyan); 3. Lateral mode (engaged green / armed cyan); 4. Approach capability (CAT I, II, III DUAL); 5. Autopilot / Flight Director engagement status. ANY MODE CHANGE is highlighted by a flashing white box for 10 seconds!
- Speed Trend Vector: A vertical magenta or green trend vector arrow on the airspeed tape. The tip indicates what the airspeed will be in exactly 10 SECONDS based on current longitudinal acceleration!

- ND Modes: 1. ROSE NAV / VOR / ILS: Full 360° compass rose centered on aircraft; 2. ARC MODE: 90° expanded forward sector, track-up oriented, displays active route, weather radar, TCAS and terrain overlays; 3. PLAN MODE: True North-up, centered on selected route waypoints, used for flight plan verification (aircraft symbol is not shown at center).

[!] EXAM TRAP (AVIATIONEXAM) - Navigation Display (ND) Radar Overlay in PLAN Mode

- CRITICAL EXAM RULE: Weather Radar (WXR) returns and Terrain (GPWS) returns CANNOT BE DISPLAYED in PLAN mode! - Reason: PLAN mode is oriented TRUE NORTH UP, whereas the weather radar antenna scans relative to the aircraft heading/track! - Weather radar overlay is available ONLY in ARC mode and ROSE mode.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): EFIS ND PLAN Mode Orientation

Rule 1: ROSE and ARC modes are oriented with aircraft Heading or Track UP and rotate in turns. Rule 2: PLAN mode is ALWAYS oriented with TRUE NORTH UP. It does not rotate. Rule 3: In PLAN mode, center of display is the selected waypoint from FMS route, not the aircraft.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under standard EASA EFIS color coding rules, what color is used on the PFD and ND to represent the active flight director guidance targets?

- [A]: Green
- [B]: Cyan
- [C]: MAGENTA
- [D]: Yellow

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Under CS-25 conventions, MAGENTA is strictly reserved for commanded flight director guidance targets, active waypoints, and FMS lateral/vertical profiles. (Green indicates engaged modes or raw data).

Q2: What does the 'Speed Trend Vector' on a modern PFD airspeed tape indicate?

- [A]: The target cruise speed
- [B]: The predicted airspeed in exactly 10 SECONDS at the current rate of acceleration or deceleration
- [C]: The maximum flap extend speed
- [D]: The windspeed component

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The speed trend vector arrow projects aircraft acceleration. The tip of the vector indicates the airspeed that will be reached in 10 seconds if current acceleration remains constant.

Q3: Why can weather radar (WXR) returns NOT be displayed on the Navigation Display when PLAN mode is selected?

- [A]: Radar transmitter powers off in PLAN mode
- [B]: PLAN mode is oriented True North up, which cannot be synchronized with the heading-referenced radar antenna scan

- [C]: Screen resolution is insufficient
- [D]: TCAS overrides radar

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The weather radar scans relative to the aircraft's physical nose (heading/track). PLAN mode shows a static map oriented True North up, making radar overlay geometrically incompatible.

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Chapter 10: Flight Management Systems (FMS & Cost Index)

Syllabus Source: CAE Oxford (pp. 349-398) | Pass Target: >= 90% AviationExam

1. FMS SYSTEM ARCHITECTURE & DUAL DATABASES

The Flight Management System (FMS) is the brain of modern navigation, integrating lateral routing (LNAV), vertical profile optimization (VNAV), and performance calculations:

- **Hardware Layout:** Consists of two redundant Flight Management Computers (FMCs) cross-talking over ARINC 429/AFDX buses, controlled via pilot Multifunction Control Display Units (MCDU) or Touchscreen Control Units.
- **Navigation Database (NDB):** A certified, READ-ONLY database updated internationally every 28 DAYS according to the ICAO AIRAC cycle. Contains worldwide airports, runways, waypoints, airways, radio aids, SIDs, STARs, and published instrument approaches. The flight crew CANNOT overwrite or alter stored airway points.
- **Performance Database (PDB):** Contains manufacturer aerodynamic drag polars, engine fuel burn curves, weight limits, and flight envelope speeds for the specific airframe/engine type.

2. COST INDEX (CI) & SPEED OPTIMIZATION

The FMS computes the most economical cruising speed and climb/descent profiles based on airline Cost Index (CI):

- **Cost Index Formula:** $\text{Cost Index} = \frac{\text{Cost of Time (\$ per hour)}}{\text{Cost of Fuel (cents per pound or kg)}}$. CI balances flight crew/maintenance hourly operational costs against fuel expenses.
- **Cost Index = 0:** Represents ZERO cost of time! The FMS commands MAXIMUM RANGE CRUISE (MRC) speed: burns minimum possible fuel per nautical mile, resulting in the slowest cruise speed and longest flight duration.
- **Cost Index = 999 (Maximum CI):** Time cost is infinitely more important than fuel! The FMS commands the maximum authorized cruising speed (V_{MO} / M_{MO}), minimizing flight time regardless of high fuel consumption.
- **Long Range Cruise (LRC):** A speed schedule flown at 99% of Maximum Range Cruise efficiency, providing ~3% to 5% higher speed for only 1% extra fuel burn.

Cost Index Value	Aeronautical Optimization Objective	Flight Performance Result
Cost Index Value	Aeronautical Optimization Objective	Flight Performance Result
CI = 0 (Minimum Fuel)	Minimum fuel consumption per nautical mile (Zero time cost).	Maximum Range Cruise (MRC). Lowest cruise speed; maximum flight time; greatest fuel economy.
Intermediate CI (Typical 20-80)	Standard airline operating balance between schedule and fuel burn.	Econ climb, Econ cruise, and Econ descent speeds calculated dynamically.
Maximum CI (e.g. 999)	Minimum flight duration (Time is critical, fuel cost ignored).	Flies at maximum certified speed (V _{MO} / M _{MO}). Highest fuel burn; shortest flight time.

[!] EXAM TRAP (AVIATIONEXAM) - Cost Index Effect on Speeds and Profiles Summary

- INCREASING Cost Index: Increases Climb Speed, Cruise Speed/Mach, and Descent Speed; steepens descent path; shortens flight time! - DECREASING Cost Index: Decreases Climb/Cruise/Descent speeds; flattens descent path; saves fuel! - CI = 0 ALWAYS corresponds to MAXIMUM RANGE (Minimum Fuel)!

3. REQUIRED NAVIGATION PERFORMANCE (RNP) VS ACTUAL (ANP)

Modern RNAV routes are certified under Performance-Based Navigation (PBN) standards:

- Required Navigation Performance (RNP): The lateral accuracy containment standard in nautical miles that the aircraft must maintain for at least 95% of total flight time (e.g. RNP 4 for oceanic, RNP 1 for terminal SIDs/STARs, RNP 0.3 for RNAV/RNP final approaches).
- Actual Navigation Performance (ANP / EPE): The current real-time estimated navigation error computed by the FMS from multi-sensor blending (GPS + DME/DME + IRS). If ANP exceeds RNP: ANP > RNP, an 'UNABLE RNP' caution illuminates on the PFD/ND, requiring an immediate missed approach or ATC notification!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Cost Index Trade-Off**

Formula: Cost Index = Time Cost (\$/hr) / Fuel Cost (cents/kg). - CI = 0: Fuel is everything -> Flies at maximum range speed (MRC) -> MINIMUM FUEL BURN. - CI = Max (99/999): Time is everything -> Flies at VMO/MMO -> MINIMUM FLIGHT TIME regardless of fuel burn.

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the operational effect of entering a Cost Index (CI) of ZERO into the Flight Management Computer?**

- [A]: The FMC commands maximum certified airspeed
- [B]: The FMC commands Maximum Range Cruise (MRC) speed, resulting in minimum fuel burn per nautical mile
- [C]: The autopilot disconnects
- [D]: The FMS switches to manual navigation

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. When CI = 0, time cost is treated as zero. The computer optimizes purely for fuel economy, flying at Maximum Range Cruise (MRC) speed.

Q2: How often is the worldwide aeronautical Navigation Database (NDB) within an aircraft FMS updated?

- [A]: Every 7 days
- [B]: Every 28 DAYS in accordance with the international ICAO AIRAC cycle
- [C]: Annually on January 1st
- [D]: Whenever the aircraft undergoes a C-check

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. International aeronautical navigation data is published and synchronized on a strict 28-day cycle known as the AIRAC (Aeronautical Information Regulation And Control) cycle.

Q3: What occurs if the Actual Navigation Performance (ANP) computed by the FMS becomes LARGER than the Required Navigation Performance (RNP)?

- [A]: The autopilot increases speed
- [B]: An 'UNABLE RNP' alert is annunciated and the flight crew must execute a missed approach or notify ATC
- [C]: The FMC switches automatically to VOR tracking
- [D]: Fuel is dumped

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. When ANP exceeds RNP, the aircraft can no longer guarantee the required obstacle clearance boundary. Flight crew procedures mandate executing an immediate go-around.

ATPL EASA STUDY GUIDE | 022 - AGK INSTRUMENTATION

Chapter 11: Stall Warning & Angle of Attack Systems

Syllabus Source: CAE Oxford (pp. 399-438) | Pass Target: >= 90% AviationExam

1. ANGLE OF ATTACK (AOA) AERODYNAMICS & SENSOR TYPES

An airfoil stalls when airflow separates from the upper surface at the Critical Angle of Attack (α_{crit} , typically 14° to 18°). An aircraft stalls at the EXACT SAME CRITICAL AOA regardless of weight, airspeed, bank angle, or altitude:

- **Pivoting Aerodynamic Vanes:** Mounted on the left and right forward fuselage. A counterweighted, aerodynamically balanced wedge vane rotates freely to align with the local oncoming relative airflow. Electrically heated to prevent ice accumulation.
- **Conical / Slotted Probes (Rosemount):** A fixed cylindrical/conical probe projecting into the airstream with two longitudinal slots. Differential pressure between the two slots is measured by a internal transducer, providing AOA measurement with no external moving parts.

2. STALL WARNING & PREVENTION DEVICES: STICK SHAKER VS STICK PUSHER

CS-25 transport certification requires clear, unmistakable aerodynamic warning prior to stall entry:

System Component	Physical Mechanism & Trigger Point	Certification Rule & Aircraft Protection
System Component	Physical Mechanism & Trigger Point	Certification Rule & Aircraft Protection
Aerodynamic Buffet	Natural turbulent wake from stalled wing root buffeting horizontal stabilizer.	Light aircraft and low-wing jets. Some swept-wing jets lack natural pre-stall buffet, requiring artificial warning devices!
Stick Shaker (Artificial Warning)	Eccentric unbalanced electric motors mounted on the control columns vibrate the yoke vigorously at ~15 to 30 Hz with a loud clattering noise.	MANDATORY CS-25 requirement: Must activate at least 5 KNOTS OR 5% ABOVE the stall speed (V_S) to allow pilot recovery before the actual aerodynamic stall.
Stick Pusher (Artificial Prevention)	Pneumatic or hydraulic actuator ram connected to the elevator control linkage. Pushes the yoke violently FORWARD with 30 to 40 kg (60-80 lb) of force!	MANDATORY on T-tail swept-wing aircraft (e.g. CRJ, bizjets) susceptible to DEEP STALL (Super-Stall) where stalled wing wake blankets the T-tail elevator, making recovery aerodynamic impossible! Pushes nose down before critical AOA.

[!] EXAM TRAP (AVIATIONEXAM) - Deep Stall (Super-Stall) & Stick Pusher Role

- Deep Stall Risk: Swept-wing aircraft with T-tail empennage and rear-fuselage engines. - Phenomenon: At high angles of attack, wingtip stall causes center of lift to shift forward, pitching the nose UP further! The turbulent separated wake of the stalled wing blankets the high T-tail elevator, completely neutralizing pitch control! - Solution: The STICK PUSHER detects rapid approach to critical AOA and forcibly drives the elevator DOWN before deep stall can ever be entered!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Stall Warning System Hierarchy

Level 1: Angle of Attack Vane detects approach to stall. Level 2: Stick Shaker vibrates control column at 1.05 Vs (tactile warning). Level 3: Stick Pusher fires at critical stall angle, pushing column forward with 35-40 kg force to prevent unrecoverable T-tail deep stall.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: Under EASA CS-25 certification requirements, at what speed margin must the artificial stall warning (stick

- [A]: Exactly at the aerodynamic stall speed
- [B]: At least 5 KNOTS or 5% ABOVE the stall speed (V_{S})
- [C]: 20 knots above stall speed
- [D]: Only when the landing gear is retracted

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CS-25 mandates that clear artificial or natural stall warning must begin at least 5 knots or 5% above the stall speed, providing sufficient reaction time for flight crew recovery.

Q2: Why are T-tail jet transport aircraft equipped with a mandatory STICK PUSHER system?

- [A]: To help the pilot trim the aircraft during take-off
- [B]: To prevent the aircraft from entering a catastrophic DEEP STALL, where the stalled wing wake blankets the T-tail, rendering elevators aerodynamically ineffective
- [C]: To retract the flaps automatically
- [D]: To prevent overspeed

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. In T-tail aircraft, wing wake washes over the elevated horizontal stabilizer at high angles of attack, causing complete loss of elevator pitch-down authority (deep stall). A stick pusher forcibly lowers the nose before this occurs.

Q3: Does an aircraft's critical stalling Angle of Attack change with increases in aircraft gross mass?

- [A]: Yes, critical AOA increases with mass
- [B]: No, the critical stalling Angle of Attack remains CONSTANT regardless of aircraft mass, bank angle, or altitude
- [C]: Yes, critical AOA decreases with mass
- [D]: It changes only with flap extension

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. An airfoil stalls when the critical angle of attack is reached (airflow separation). While the stall SPEED increases with mass and load factor, the critical stalling ANGLE OF ATTACK is aerodynamically constant.

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Chapter 12: Ground Proximity Warning Systems (GPWS & EGPWS)

Syllabus Source: CAE Oxford (pp. 439-488) | Pass Target: >= 90% AviationExam

1. CLASSIC GPWS OPERATING MODES (MODES 1 TO 7)

Classic GPWS warns of impending Controlled Flight Into Terrain (CFIT). It relies on Radio Altimeter rate of change and aircraft configuration from 50 ft to 2,500 ft AGL:

GPWS Warning Mode	Aeronautical Trigger Condition	Cockpit Visual & Synthetic Aural Warnings
Mode 1: Excessive Sink Rate	Radio altitude sinking excessively fast.	'SINK RATE, SINK RATE' (amber); if uncorrected: 'PULL UP' (red light + continuous aural siren).
Mode 2: Terrain Closure Rate	Rapidly rising terrain beneath aircraft.	'TERRAIN, TERRAIN' (amber); if closure rate exceeds boundary: 'PULL UP, PULL UP' (red).
Mode 3: Altitude Loss After Take-Off	Barometric altitude sink after take-off or go-around before reaching 1,500 ft AGL.	'DON'T SINK, DON'T SINK' (amber voice alert repeated until positive climb established).
Mode 4: Unsafe Terrain Clearance	Landing gear or flaps NOT in landing position at low radio altitude.	Mode 4A (Gear up, speed < 190 kt): 'TOO LOW, GEAR'. Mode 4B (Flaps up, speed < 159 kt): 'TOO LOW, FLAPS'. At higher speeds: 'TOO LOW, TERRAIN'.
Mode 5: Below Glideslope	Descent more than 1.3 dots below ILS glideslope beam.	'GLIDESLOPE, GLIDESLOPE' (amber alert; volume increases as glideslope deviation deepens).
Mode 6: Bank Angle / Callouts	Excessive bank angle (> 35° to 45° depending on altitude); plus advisory altitude callouts.	'BANK ANGLE, BANK ANGLE'; plus synthetic height calls ('ONE THOUSAND', 'FIVE HUNDRED', 'MINIMUMS').
Mode 7: Windshear Detection	Reactive windshear detected via accelerometer and air data comparison.	Red WINDSHEAR light on PFD + two-tone siren + synthetic voice 'WINDSHEAR, WINDSHEAR, WINDSHEAR'.

2. ENHANCED GPWS (EGPWS / TAWS): FORWARD LOOKING TERRAIN (FLTA)

Classic GPWS had a deadly limitation: its radio altimeter looked ONLY straight down, providing zero warning when flying toward a vertical cliff or rising mountain ridge (CFIT risk):

- EGPWS / TAWS Architecture: Adds a worldwide digital terrain elevation database, obstacle database (towers, bridges), and high-precision GPS aircraft positioning.
- Forward Looking Terrain Avoidance (FLTA): Projects a virtual look-ahead caution envelope (40 to 60 seconds ahead) and warning envelope (20 to 30 seconds ahead) along the aircraft's projected flight path. Issues alerts BEFORE the aircraft arrives at rising terrain!
- Premature Descent Alert (PDA): Warns if aircraft descends significantly below the normal 3° descent profile on approach prior to reaching the runway.
- ND Terrain Display Colors: Solid Red: Terrain > 2,000 ft above aircraft; Solid Yellow: Terrain 500 ft below to 2,000 ft above; Green: Terrain 2,000 ft to 500 ft below aircraft altitude.

[!] EXAM TRAP (AVIATIONEXAM) - GPWS / EGPWS 'PULL UP' Emergency Flight Maneuver

Upon receiving any RED 'PULL UP' or 'TERRAIN AHEAD PULL UP' warning in IMC or night: - Disconnect Autopilot and Autothrottle immediately! - Apply MAXIMUM TOGA THRUST! - Roll wings level and pitch up smoothly to the PITCH LIMIT INDICATOR (PLI) or stick shaker angle! - DO NOT RETRACT GEAR OR FLAPS until terrain clearance is fully assured!

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): GPWS PULL UP Escape Maneuver**

Trigger: 'PULL UP' or 'TERRAIN, PULL UP' warning. Step 1: Disconnect Autopilot and Autothrottle immediately. Step 2: Advance thrust levers to maximum TOGA thrust. Step 3: Rotate smoothly to pitch limit indicator / stick shaker margin. Step 4: Keep wings level. DO NOT change gear or flap configuration until safe climb is verified.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: What is the primary operational advantage of an Enhanced GPWS (EGPWS / TAWS) over a conventional GPWS?**

- [A]: It measures cabin differential pressure
- [B]: It provides Forward Looking Terrain Avoidance (FLTA) using a worldwide terrain database and GPS, detecting rising terrain ahead before flying over it
- [C]: It replaces the weather radar
- [D]: It operates only in VMC

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Classic GPWS senses terrain directly downward via radio altimeter, creating blind spots against steep cliffs. EGPWS compares GPS position against a digital terrain database to warn 40-60 seconds ahead of impact.

Q2: Which GPWS mode triggers the cockpit voice warning 'DON'T SINK, DON'T SINK'?

- [A]: Mode 1: Excessive descent rate
- [B]: Mode 2: Rapid terrain closure
- [C]: Mode 3: Inadvertent altitude loss after take-off or go-around
- [D]: Mode 5: Below glideslope

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [C]. Mode 3 monitors barometric altitude loss following take-off or missed approach with gear/flaps not in landing configuration, alerting 'DON'T SINK'.

Q3: When the flight crew receives an authentic EGPWS 'PULL UP' warning in IMC, why must the landing gear be retracted?

- [A]: Gear retraction increases parasitic drag
- [B]: Retracting gear/flaps causes a momentary loss of lift (sink) and alters aircraft trim, which could result in ground impact
- [C]: Regulations prohibit gear movement
- [D]: The warning will cancel

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Standard CFIT escape procedures mandate leaving configuration untouched during initial pull-up to avoid aerodynamic settling or secondary stall while achieving maximum climb angle.

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Chapter 13: Traffic Alert & Collision Avoidance (TCAS II v7.1)

Syllabus Source: CAE Oxford (pp. 489-538) | Pass Target: >= 90% AviationExam

1. TCAS II ARCHITECTURE & THE TAU (TAU) PRINCIPLE

TCAS II operates independently of ground Air Traffic Control radar, interrogating SSR Mode C and Mode S transponders on 1030 MHz and receiving replies on 1090 MHz within a ~30 to 50 NM surveillance range:

- Tau Concept (Time to CPA): TCAS does NOT calculate collision risk based purely on physical distance! Instead, it evaluates TAU (Tau): Time to Closest Point of Approach (CPA): $\text{Tau} = \text{Range} / \text{Closure Rate}$: $\text{Tau} = r / (-r_{\text{dot}})$.
- Threat Classification Hierarchy: 1. Non-Threat Traffic: Open white diamond (> 6 NM or > 1,200 ft vertical); 2. Proximity Traffic: Solid white diamond (within 6 NM and 1,200 ft, but non-converging); 3. Traffic Advisory (TA): Solid amber circle; 4. Resolution Advisory (RA): Solid red square.

Alert Level	Tau Time Threshold (Cruise)	Cockpit Visual & Synthetic Aural Warning	Mandatory Pilot Action
Alert Level	Tau Time Threshold (Cruise)	Cockpit Visual & Synthetic Aural Warning	Mandatory Pilot Action
Traffic Advisory (TA)	Tau = 35 to 48 Seconds	Solid AMBER circle on ND. Synthetic voice: 'TRAFFIC, TRAFFIC'.	Attempt visual acquisition of conflicting traffic. PREPARE for possible Resolution Advisory. DO NOT MANEUVER solely based on TA!
Resolution Advisory (RA)	Tau = 20 to 35 Seconds	Solid RED square on ND. Pitch cues on PFD vertical speed tape (green fly-to arc; red avoid arc). Aural: 'CLIMB', 'DESCEND', 'LEVEL OFF'.	FLY THE RA IMMEDIATELY within 5 SECONDS! TCAS RA commands take strict priority over all ATC clearances!

2. TCAS II VERSION 7.1 ENHANCEMENTS & SENSE REVERSALS

Following the fatal 2002 Überlingen mid-air collision, TCAS Version 7.1 introduced vital safety modifications:

- 'LEVEL OFF' Aural Replaces 'Adjust Vertical Speed': In initial RA encounters where traffic levels off, TCAS v7.1 now commands 'LEVEL OFF, LEVEL OFF', instructing pilots to reduce vertical speed to zero fpm smoothly, preventing excessive climb/descent.
- Coordinated Sense Reversals: When two TCAS II aircraft encounter each other, their Mode S transponders negotiate resolution senses automatically: one commands CLIMB while the other commands DESCEND. If one pilot fails to comply (flying opposite to RA), TCAS v7.1 detects the non-compliance and issues an immediate SENSE REVERSAL ('CLIMB, CLIMB NOW!' or 'DESCEND, DESCEND NOW!'). Reaction time for reversal RA: 2.5 SECONDS!

[!] EXAM TRAP (AVIATIONEXAM) - ATC vs TCAS RA Absolute Legal Priority

- STRICT EASA / ICAO LAW: A TCAS Resolution Advisory (RA) ALWAYS TAKES PRECEDENCE over conflicting ATC instructions! - If ATC says 'Descend FL 350' but TCAS commands 'CLIMB, CLIMB': - YOU MUST CLIMB! Never follow an ATC clearance that contradicts a TCAS RA! - Notify ATC as soon as workload permits: '[Callsign], TCAS RA'.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)**[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): TCAS II Resolution Advisory Acti**

Rule 1: Disconnect autopilot and execute RA within 5 SECONDS applying 0.25g. Rule 2: OBEY TCAS OVER ATC: If controller orders opposite maneuver, pilot MUST follow TCAS RA. Rule 3: TCAS II commands ONLY VERTICAL maneuvers (Climb/Descend). It NEVER commands turns. Rule 4: Call ATC: '[Callsign], TCAS RA'.

3. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)**Q1: If an air traffic controller issues a radar descent clearance that directly contradicts a TCAS Resolution Adv**

- [A]: Follow the ATC instruction because ATC has radar separation overview
- [B]: Follow the TCAS RA immediately, disengaging autopilot if necessary, and advise ATC as soon as practical
- [C]: Level off and ask ATC to confirm
- [D]: Turn 90 degrees right

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Under international aviation regulations, a TCAS RA takes absolute legal precedence over conflicting ATC clearances. Pilots must immediately respond to the RA within 5 seconds and notify ATC.

Q2: How does TCAS II calculate the time to Closest Point of Approach (CPA) to trigger a Traffic Advisory or Re

- [A]: By measuring barometric pressure difference
- [B]: Using the Tau (Tau) formula: Range divided by Closure Rate (r / r_{dot})
- [C]: By checking the GPS ground distance
- [D]: Using the aircraft Mach number

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. TCAS II evaluates collision risk using Tau (time to CPA), which is calculated as instantaneous slant range divided by closing velocity. This accounts for high closing speeds in head-on traffic.

Q3: What critical safety change was implemented in TCAS II Version 7.1 to improve pilot compliance during ve

- [A]: Added horizontal steering commands
- [B]: Replaced 'Adjust Vertical Speed, Adjust' with the clear synthetic voice command 'LEVEL OFF, LEVEL OFF'
- [C]: Increased surveillance range to 200 NM
- [D]: Inhibited all warnings below 10,000 ft

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. TCAS v7.1 replaced the confusing 'Adjust Vertical Speed' command with 'LEVEL OFF, LEVEL OFF' to guide pilots smoothly to zero vertical speed, preventing overshoot into conflicting flight levels.

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Chapter 14: Automatic Flight Control & Autoland (CAT I/II/III)

Syllabus Source: CAE Oxford (pp. 539-600) | Pass Target: >= 90% AviationExam

1. AUTOMATIC FLIGHT CONTROL SYSTEM (AFCS) HIERARCHY

The AFCS integrates the Autopilot (AP), Flight Director (FD), and Autothrottle / Autothrust (A/T) into an automated flight envelope:

- Flight Director (FD): Computes steering commands displayed as crosshairs or single-cue command bars on the PFD. Indicates where the pilot MUST steer the aircraft manually to intercept and maintain the selected flight path.
- Inner Loop (Stability Augmentation / Fast Loop): Damps aerodynamic oscillations and holds attitude. Senses angular rates from rate gyros/accelerometers and deflects control surfaces (e.g. Yaw Damper to eliminate Dutch Roll). Operates with rapid control surface movements.
- Outer Loop (Guidance / Slow Loop): Tracks navigation paths and targets: Altitude Hold, Heading Select, VOR/LOC, LNAV, VNAV, Glide Path. Displaces control surfaces slowly to guide aircraft along flight trajectory.
- Control Wheel Steering (CWS): A semi-automatic mode where pilot applies manual force on control yoke; autopilot follows pilot input. When yoke pressure is released, autopilot locks onto and maintains the new attitude!

2. ICAO / CS-AWO PRECISION LANDING CATEGORIES

All-Weather Operations (CS-AWO) categorize instrument approaches by Decision Height (DH) and Runway Visual Range (RVR):

Approach Category	Minimum Decision Height (DH)	Minimum Runway Visual Range (RVR)	Autoland System Architecture
CAT I	DH >= 200 ft (60 m)	RVR >= 550 m (or 800 m vis)	Standard single autopilot or manual flight with Flight Director.
CAT II	100 ft <= DH < 200 ft	RVR >= 300 m	Fail-Passive autopilot or dual flight directors. Radio altimeter mandatory.
CAT IIIa	DH < 100 ft or NO DH	RVR >= 175 m	Fail-Passive or Fail-Operational autoland system with automatic rollout/manual rollout.
CAT IIIb	DH < 50 ft or NO DH	50 m <= RVR < 175 m	Fail-Operational autoland system with automatic rollout guidance to runway center.
CAT IIIc	NO DH (Zero DH)	NO RVR (Zero RVR)	Zero-zero landing and taxi guidance. (Not commercially certified due to ground rescue limits).

3. FAIL-PASSIVE VS FAIL-OPERATIONAL AUTOLAND ARCHITECTURE

Transport category aircraft require certified multi-channel redundancy for low-visibility autolandings:

- **Fail-Passive System (Single-Failure Tolerant):** In the event of an autopilot channel failure below alert height, NO OUT-OF-TRIM condition occurs, but the autopilot DISCONNECTS automatically! The pilot must immediately take manual control and execute a missed approach (go-around) if visual reference is not established. Typical for CAT II and CAT IIIa.
- **Fail-Operational System (Dual-Failure Tolerant):** Features at least two independent autopilot channels operating simultaneously (duplex or triplex). If one channel fails below the Alert Height (typically 100-200 ft), the remaining channel continues the approach, flare, touchdown, and rollout AUTOMATICALLY with NO PILOT INTERVENTION! Mandatory for CAT IIIb.

[!] EXAM TRAP (AVIATIONEXAM) - Autoland Flare Mode Trigger & Transition

- Below 50 ft AGL: Sensed by the Radio Altimeter! - The autopilot enters FLARE mode: Pitch attitude increases smoothly to reduce sink rate to ~1-2 ft/sec. - Autothrottle automatically retards thrust levers to IDLE (FMA annunciates 'RETARD' at ~30-20 ft). - At touchdown: Rollout mode tracks the ILS localizer centerline via nosewheel steering and rudder.

STEP-BY-STEP PROBLEM SOLVING (FOR DUMMIES / PASO A PASO)

[+] STEP-BY-STEP ALGORITHM / PASO A PASO - Step-by-Step Algorithm (Dummies Guide): Fail-Passive vs Fail-Operational A

Rule 1 (Fail-Passive): Single failure below alert height -> Autopilot disconnects in neutral trim with zero out-of-trim transients. Pilot must see runway and land manually or go around. Rule 2 (Fail-Operational): Single failure below alert height -> Autopilot continues approach, flare, and rollout automatically without pilot intervention (Cat IIIb/IIIc).

4. AVIATIONEXAM PRACTICE QUESTIONS (WITH EXPLANATIONS)

Q1: In an Automatic Flight Control System (AFCS), what is the difference between an 'inner loop' and an 'outer loop'?

- [A]: Inner loop controls engines, outer loop controls flaps
- [B]: Inner loop provides rate damping and short-term stability augmentation; outer loop provides long-term path guidance (altitude hold, heading, nav tracking)
- [C]: Inner loop is manual, outer loop is automatic
- [D]: Inner loop operates only on the ground

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. The inner loop stabilizes the aircraft about its axes using rate gyros (damping Dutch roll and pitch oscillations). The outer loop guides the aircraft along selected navigation paths (holding altitude or tracking localizer).

Q2: What is the defining operational characteristic of a 'Fail-Operational' autoland flight guidance system?

- [A]: Following an autopilot failure, the aircraft automatically executes a go-around
- [B]: Following a single system failure below alert height, the approach, flare, and landing can be completed automatically by the remaining operating channel without pilot intervention
- [C]: It requires manual landing flare

- [D]: It operates on battery power only

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. Fail-operational systems feature multiple redundant channels. If one channel fails below alert height, the remaining channel continues and lands the aircraft automatically with zero interruption.

Q3: What is the minimum Runway Visual Range (RVR) required for an ICAO Category II (CAT II) precision ILS a

- [A]: 550 m
- [B]: 300 m
- [C]: 175 m
- [D]: 75 m

[?] OFFICIAL RATIONALE & ANALYSIS - Detailed Answer Explanation

CORRECT: [B]. CAT II requires a minimum Decision Height between 100 ft and 200 ft, and an RVR of not less than 300 meters.